Choptuik's echo period at large dimension: evidence for a finite limit $\Delta_\infty = 1.64389375$, and selection beyond all orders in $1/D$

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We compute the echo period $\Delta(D)$ of Choptuik's discretely self-similar critical solution of massless scalar collapse in D spacetime dimensions at 857 points $8.25 \le D \le 61.03$ with a double-double (about 32-digit) solver, consistent with smooth curves to 4×10^{-15}, and with an independent double-precision solver for $4 \le D \le 40$. One rational function of $1/(D-2)$ reproduces Δ and the critical exponent γ for all $D \ge 4$ within the precision of the points. Series in $1/D$ fitted on $D \le 45$ predict the points up to $D = 60.1$ to 5×10^{-13} (to $D = 61$: 7×10^{-13}), 40 to 300 times better than series in half-integer powers; this supports, but does not prove, analyticity in $1/D$, which then gives $\Delta_\infty = 1.64389375 \pm 1.5 \times 10^{-8}$ (the spread over 48 fit choices); the critical exponent, computed in double-double up to $D = 61$, tends to $\gamma_\infty = 1/2$ to about 5×10^{-7}. We explain this limit, given the conjectured selection rule below and the uniqueness of the unstable mode: at $D = \infty$ an unstable mode with $\lambda = 1/\gamma = 2$ is generated by an exact symmetry, a shift of r^2, of the leading-order large-D equations of Rozali and Way. A power-law approach to $\Delta = 0$ with exponent at least 0.01, as conjectured in the literature, is excluded by the data; a logarithmic one is disfavoured. We show that, within the scaling measured on the solutions, the $1/D$ expansion of the self-similar field equations can be carried to all orders for any period and any time profile, so the period (and likewise γ) is selected beyond all orders in $1/D$. Without gravity the obstruction is a closed-form light-cone coefficient, which we identify as the Stokes multiplier of a mode born at large self-similar radius. We conjecture that the selection rule at $D = \infty$ is the vanishing of the Stokes multipliers of an n-free inner problem; its higher-harmonic conditions give the shape of the echo, $|F_3/F_1| = 0.0551$, in agreement with the data. Truncated to three harmonics, the full set of conditions selects $\Delta = 1.619$, 1.5% below Δ_∞, and with five harmonics $\Delta = 1.64516$, 0.077% above (one harmonic: 29% above). Without truncation, an amplitude expansion to twelve orders selects $\omega = (1.0000 \pm 0.0001)\,\omega_\infty$ (the range of several estimators, not an error bound) at amplitude 2.720 (measured: 2.70–2.72). The conjectured rule thus gives $\Delta_\infty = 1.6439 \pm 0.0002$ from the $D = \infty$ equations alone, against 1.64389375 from the solutions; the rule itself is not yet proved. The computations, derivations and text were produced by an AI model (Claude, Anthropic) under the author's direction and have not yet been reviewed by a human expert (Sec. VII).

I. INTRODUCTION

Fine-tuning a one-parameter family of initial data for a massless scalar field to the threshold of black-hole formation, Choptuik found that the evolution approaches a universal critical solution that repeats itself on ever smaller scales: it is discretely self-similar (DSS) with echo period Δ, and black holes formed just above threshold have radius $r_{\rm BH} \propto (p - p^*)^\gamma$ [1]; in D dimensions their mass scales as $M \propto (p - p^*)^{(D-3)\gamma}$, which in four dimensions is the familiar $M \propto (p - p^*)^\gamma$. The scaling follows from the critical solution having a single unstable mode, with $\gamma = 1/\lambda$ for its growth rate λ [7]. Throughout, γ is this radius exponent. In four dimensions $\Delta = 3.445452402(3)$ [10] and $\gamma \simeq 0.374$ [9]; see the review [11].

The dependence on the spacetime dimension D has been studied numerically, mostly at percent-level precision, for $3.5 \le D \le 14$ (mostly integer D) [2, 12–16], and to about 10^{-6} for continuous $3.05 < D < 5.5$ by Ecker, Ecker, Grumiller and Jechtl [17], who find a maximum of Δ near $D = 3.756$. Ecker, Ecker and Grumiller [18] constructed analytic DSS solutions at large D and, with [17], conjectured $\Delta \to 0$ as $D \to \infty$; they also report the expectation $\gamma \to 1/2$, which goes back to Rozali and Way [8]. Rozali and Way derived the leading-order large-D equations of a self-gravitating scalar field, observed that they are parabolic with the roles of space and time swapped, and estimated $\gamma = 1/2$ from the divergence of their oscillating solutions. Clark and Pimentel [3] constructed continuously self-similar critical solutions at large D by matched asymptotics. Large D has been a fruitful limit in gravity [19].

This paper reports the following.

1. $\Delta(D)$ at 857 points $8.25 \le D \le 61.03$ from a double-double solver, and Δ, γ for $4 \le D \le 40$ from a double-precision solver; one rational formula in $1/(D-2)$ for all $D \ge 4$ (Sec. III). It reproduces the published values

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at $D = 4$ and the maximum of Δ below $D = 4$; at $D = 5$ it differs from [17] by 6×10^{-5}, a difference that the published higher-resolution data of [17] trace to resolution.

2. Held-out tests that support analyticity of Δ in $1/D$, giving $\Delta_\infty = 1.64389375 \pm 1.5 \times 10^{-8}$; a power-law approach to zero is excluded and a logarithmic one disfavoured; given analyticity, nine constants tested as closed forms for Δ_∞ are excluded (Sec. IV).

3. The $1/D$ expansion never fixes the period (Proposition 1); the leading-order problem and its light-cone coefficient in closed form (Proposition 2); what gravity does to the non-analytic piece (Proposition 3); and a proof that the late orders of the $1/D$ series encode that coefficient as a Stokes multiplier (Proposition 4) (Sec. V); and an exact unstable mode with $\lambda = 2$ at $D = \infty$, from a shift symmetry of r^2 of Rozali and Way's leading-order equations, which explains $\gamma_\infty = 1/2$ under the selection conjecture and the uniqueness of the unstable mode (Proposition 5).

4. An n-free inner problem whose Stokes multipliers, by a conjecture stated with its hypotheses and the points where a proof is missing (Conjecture 1), select the period; its higher-harmonic conditions give $|F_3/F_1| = 0.0551$ and $|F_5/F_1| = 0.00314$ for the echo profile at $D = \infty$, in agreement with 0.0549–0.0552 and 0.0032–0.0034 extrapolated from the solutions (Sec. V F). The harmonic-1 condition, which would give Δ_∞, is not yet settled: truncated to three harmonics, the conditions select $\omega = 1.0156\,\omega_\infty$ with the measured shape, with five about $0.99923\,\omega_\infty$, and without truncation, at twelve orders, $(1.0000 \pm 0.0001)\,\omega_\infty$ at amplitude 2.72 (measured: 2.70–2.72).

5. An analysis of the next-to-leading-order condition of [18], which within this framework does not select Δ (Sec. V G).

The inner equations of Sec. V F are, in other variables, the leading-order large-D system of Rozali and Way [8], who also noted its heat-equation character and estimated $\gamma = 1/2$; what is new here is their use for the discretely self-similar solution: the selection rule and the symmetry argument of Proposition 5.

II. EQUATIONS AND NUMERICAL METHOD

We use the self-similar radius x (radius over time to the singularity, normalized so that the past light cone of the singularity is at $x = 1$) and $\tau = -\ln(-t)$, or the phase $\theta = \omega\tau$ with $\omega = 2\pi/\Delta$. With $n = D - 2$, p and m the (scaled) derivatives of the scalar field along the two radial null directions, a the radial metric function ($b = a^2$) and c the lapse-type function that sets the light cones, the equations are

$$x c_x = (n-1)\,c\,(b-1), \qquad x b_x = b\,(p^2+m^2) - (n-1)\,b\,(b-1), \tag{1}$$

$$x(x-c)\,p_x = -\omega x\,p_\theta + Gp - cp + \tfrac{n}{2}\,c\,(p+m), \qquad x(x+c)\,m_x = -\omega x\,m_\theta - Gm + cm - \tfrac{n}{2}\,c\,(p+m), \tag{2}$$

with $G = (n-1)\,c\,(b-1)$, on $0 \le x \le 1$; p, m contain only odd harmonics of θ and b, c only even ones; the solution is regular at $x = 0$, analytic at the past light cone $x = 1$, and $c(1) = 1$ fixes the gauge.

Double-precision solver ($4 \le D \le 40$): spectral multiple shooting in x with Fourier modes in τ (23–47 modes), with the integration-step error extrapolated out; near $D = 4$ the points are extrapolated in the mode count. Above $D \approx 26$ the period is selected by an effect that weakens exponentially with D, and rounding at 10^{-16} moves Δ by 10^{-7}–10^{-6} at $D = 30$–40.

Double-double solver ($8.25 \le D \le 61.03$): double-double arithmetic (about 32 digits) on SIMD vectors; Taylor series in x of order 36 with truncation 10^{-32}; centre and horizon series to 10^{-33}; collocation in τ on the half period (32 points); an exact Jacobian by forward-mode duals with a double-double LU; continuation in D with deflated Newton steps. The Jacobian has a ladder of near-null directions (high τ-harmonics of the centre data, singular values falling roughly like $e^{-0.7k}$); from $D = 35$ the centre harmonics above 13 are pinned, which moves Δ by 5×10^{-15} at $D = 35$ and 3×10^{-17} at $D = 40$. Checks are listed in Table I. The 857 points fit smooth curves to about 4×10^{-15}; the estimated error per point is about 10^{-16} to $D = 60.1$ and a few 10^{-15} above, where the tolerance was relaxed.

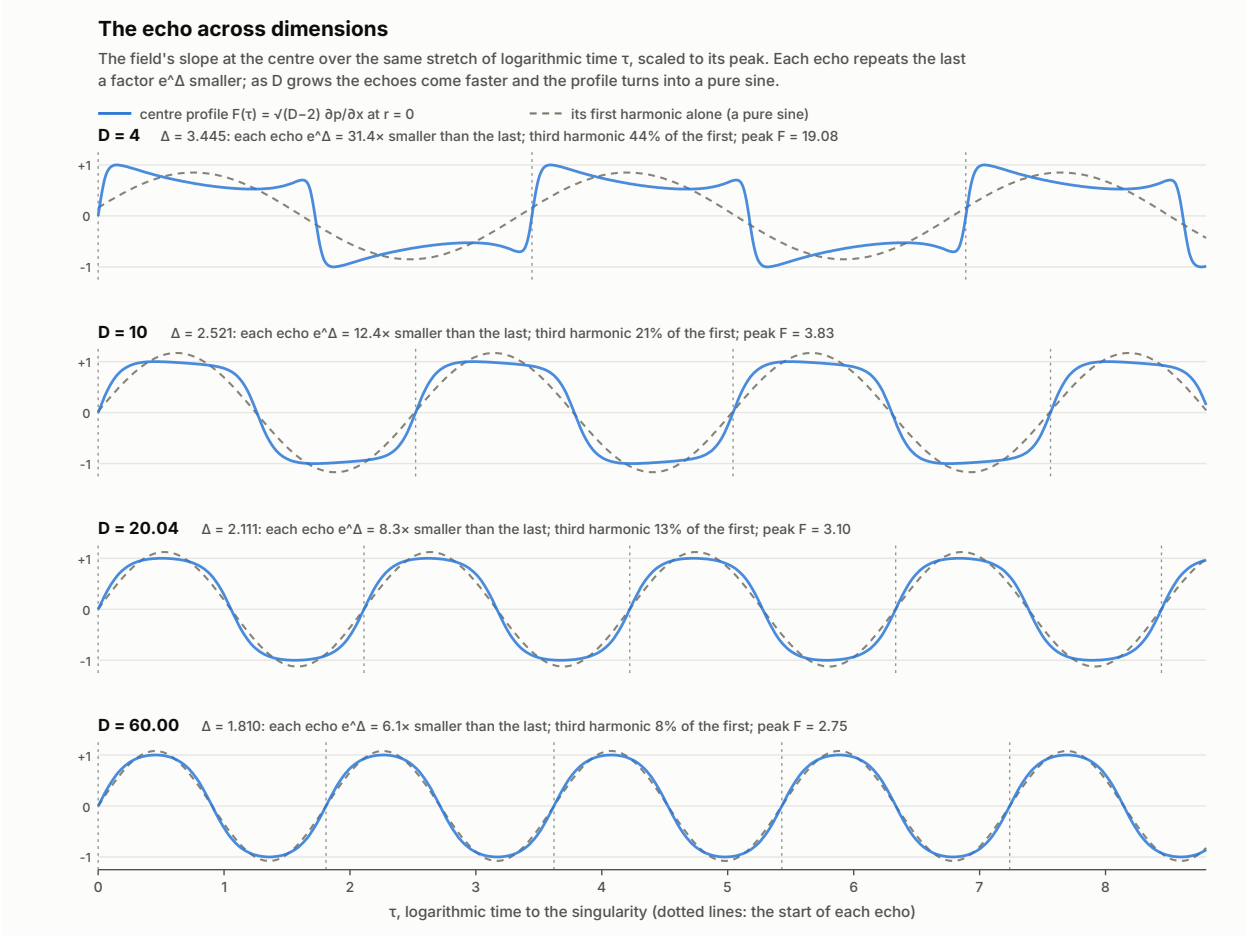


FIG. 1. The echo across dimensions: the centre slope of the field, $F(\tau) = \sqrt{D-2}\,\partial_x p$ at $r = 0$, over the same stretch of logarithmic time τ for $D = 4$, 10, 20 and 60, each scaled to its peak (dashed: its first harmonic). At $D = 4$ the profile is close to a square wave and each echo is $e^{\Delta} = 31\times$ smaller than the last; at $D = 60$ it is nearly a pure sine and the factor is 6.1. The $D = 4$ profile is from the double-precision solver (39 harmonics); the others are double-double solutions.

TABLE I. Checks of the double-double solver.

Check	Result
double-double arithmetic against mpmath	1.3–1.6×10^{-32} relative
horizon series against 40-digit mpmath	2×10^{-32}
Newton at $D = 8$	quadratic, $\lvert F\rvert = 5 \times 10^{-31}$ in 4 steps
$\Delta(7)$, $\Delta(7.75)$ against 39-mode references ($\pm 3 \times 10^{-11}$)	$+3.1 \times 10^{-12}$, -2.8×10^{-12}
32 / 48 / 64 collocation points in τ, $D = 7$–17	Δ agrees to 10^{-16}
32 / 48 collocation points, $D = 40$	9×10^{-22}
harmonics pinned above 13 / above 15, $D = 40$	3×10^{-17}
second machine (separate build), $D = 59.84$–60.05	2–4×10^{-16}
32 points $D = 60.95$–61.03 against the $1/D$ fit on $D \le 60.1$	2.6×10^{-15}

III. RESULTS

A. One formula for all $D \geq 4$

With $\varepsilon = 1/n = 1/(D-2)$,

$$\Delta(D) = \frac{\sum_{k=0}^{8} a_k \varepsilon^k}{1 + b_1\varepsilon + b_2\varepsilon^2}, \qquad \gamma(D) = \frac{\frac{1}{2} + \sum_{k=1}^{7} e_k \varepsilon^k}{1 + f_1\varepsilon + f_2\varepsilon^2}, \tag{3}$$

TABLE II. Coefficients of Eq. (3).

k	a_k	b_k	e_k	f_k
0	$+1.6438975180$	1	$+0.5$ (fixed)	1
1	$+4.0378045077 \times 10^{1}$	$+1.8251453626 \times 10^{1}$	$+1.8083629635 \times 10^{1}$	$+3.7683686025 \times 10^{1}$
2	$+2.5233285989 \times 10^{2}$	$+6.8182050167 \times 10^{1}$	$+1.5241588758 \times 10^{2}$	$+3.4245505235 \times 10^{2}$
3	$+2.0673502321 \times 10^{2}$		$-5.2341588541 \times 10^{1}$	
4	$-3.5709850575 \times 10^{2}$		$+9.9259594505$	
5	$+3.1347320996 \times 10^{2}$		$-9.8386450353 \times 10^{1}$	
6	$-4.1891374574 \times 10^{2}$		$+7.4529506074 \times 10^{1}$	
7	$+3.2695252782 \times 10^{2}$		$-4.8272358903 \times 10^{1}$	
8	$-1.6050467699 \times 10^{2}$			

TABLE III. Values. Δ for $D = 10$–60 is double-double (interpolated to 10^{-14} where D is not a computed point); $\Delta(8)$ is a separate double-double run; Δ for $D = 4$–7 and γ to $D = 40$ are double-precision; $\gamma(50)$, $\gamma(100)$, $\gamma(1000)$ are formula values. $\Delta(70)$ and $\Delta(100)$ are extrapolations on which all models fitting the double-double points agree (to 10^{-11}, 2×10^{-8}); $D = 1000$ and ∞ assume analyticity in $1/D$.

D	Δ	γ	D	Δ	γ
4	3.4454524	0.3739610	20	2.112080154580	0.4737107
5	3.2217019	0.4131269	30	1.964957899258	0.4807927
6	3.0094832	0.4300795	40	1.888568197910	0.484776
7	2.8427532	0.4399912	50	1.841659503007	0.48736
8	2.711790054444	0.4467164	60	1.809889459975	
10	2.521078919562	0.4555879	70	1.78693476494	
12	2.389188553641	0.4614056	100	1.7450264844	0.49313
15	2.253098407194	0.4673656	1000	1.65424086	0.4992
			∞	$1.64389375 \pm 1.5 \times 10^{-8}$	1/2 (assumed)

with the coefficients of Table II, fitted on the double-precision points $4 \le D \le 30.2$ (so a_0 is not Δ_∞: it lies 4×10^{-6} above the double-double limit of Sec. IV). It is an empirical fit; no earlier formula for $\Delta(D)$ was found, and the earlier fit for γ, $0.466\,(1 - e^{-0.408D})$ on $3.5 \le D \le 14$ [14], has a different form and limit. They reproduce Δ to 5×10^{-10} ($4 \le D \le 4.3$), 4×10^{-9} (4.3–20) and 10^{-8} (20–30), and γ to 3.4×10^{-7}, 3.3×10^{-8} and 2.3×10^{-9}, each within the precision of the points. Against the double-double points, which the fit never saw, the formula holds to 2.7×10^{-9} on 8.25–21, 7.5×10^{-9} to $D = 30$ and 5.8×10^{-8} to $D = 61$ (Fig. 2). The limit $\gamma_\infty = 1/2$ is imposed in the γ fit, not derived; unconstrained fits of the double-precision γ on $D \ge 8$–12 give 0.4996–0.4998. The critical exponent computed in double-double (the root of the determinant of the exact growth-mode matrix, 28 points for $D = 10$–61, accurate to about 10^{-12} for $D = 16$–58) pins the limit much more tightly: in free fits in powers of $1/n$, $\lambda_\infty - 2$ ($\lambda = 1/\gamma$) falls by a factor of about 3.7 with each added power, to 10^{-6} at five powers on $D \ge 30$, so $\gamma_\infty = 1/2$ to about 5×10^{-7}, with $\gamma = 1/2 - 0.7581/n + \ldots$. The denominators vanish only at $D = -3.2, -11.0$ (Δ) and $D = -13.3, -20.4$ (γ). Values are listed in Table III.

B. Comparison with published values

At $D = 4$ the formula gives $\Delta = 3.4454524$ and $\gamma = 0.3739610$, in agreement with $\Delta = 3.445452402(3)$ [10] to about 10^{-8} and with $\gamma = 0.373961 \pm 10^{-6}$ [17] to about 10^{-7}. Below $D = 4$ a cubic through the 47-mode double-precision points at $D = 3.70$–3.80 puts the maximum of Δ at $D = 3.75570$, $\Delta = 3.466774$, in agreement to 2×10^{-6} with $D = 3.755726$, $\Delta = 3.466772$ of [17].

At $D = 5$, Ref. [17] gives $\Delta = 3.22176$ and $\gamma = 0.41322$, which are 5.8×10^{-5} and 9.3×10^{-5} above our values. Our $\Delta(5) = 3.2217019446$ comes from two independent codes: the double-precision solver at 31, 39 and 47 Fourier modes ($3.221701945 \pm 7 \times 10^{-11}$ extrapolated) and the double-double solver at $M = 48$ and 64 collocation points (3.22170194461806, 3.22170194462672), which agree to 10^{-11}. The published data of [17] (repository CriticalSpacetimeCrystal) show their $D = 5$ value still moving by -1.66×10^{-4} between 256 and 512 modes, toward ours, in the range $D \gtrsim 4.8$ where they note that their light-cone expansion order must be increased; their offset from our values grows smoothly from 2.7×10^{-7} at $D = 4$ (within their stated $\pm 10^{-6}$) to 5×10^{-4} at $D = 5.5$, and at $D = 3.7$ the two agree to 8×10^{-8}, so there is no difference of convention. Their γ list has no points for $4.79 < D < 5.49$, where they report that their perturbation method fails, so their $\gamma(5)$ is not a direct computation in the published data;

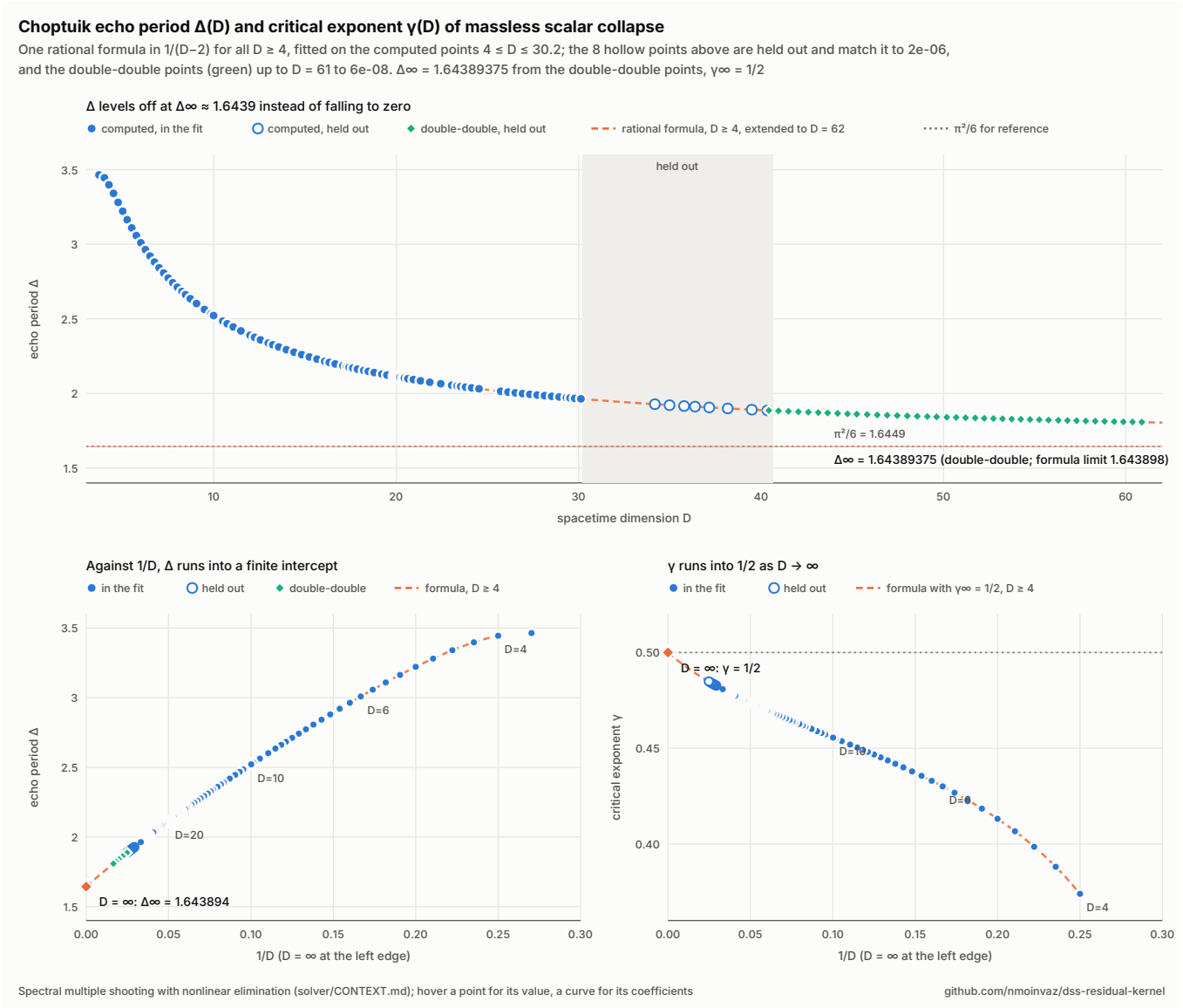


FIG. 2. $\Delta(D)$ and $\gamma(D)$ with the rational formula (3) (dashed), fitted on $4 \le D \le 30.2$; hollow points are held out, and green diamonds are double-double points up to $D = 61$.

ours, 0.41312694 (stable to 10^{-7} across resolutions), agrees with Bland and Kunstatter's 0.4131 ± 0.0001 [15]. Below $D = 4$, γ extrapolated from our 47-mode points suggests $\gamma(3.5) \lesssim 0.33$, against 0.349 ± 0.003 in [14]; our points there are under-resolved, and this is an open check.

C. Physical reading

Each echo is a factor e^{Δ} smaller than the last: 31 at $D = 4$, 12 at $D = 10$, 5.2 at $D = \infty$. A near-critical collapse makes about $\gamma \ln(1/|p - p^*|)/\Delta$ echoes, so each further echo costs $\Delta/(\gamma \ln 10)$ decimal digits of fine-tuning: 4.0 at $D = 4$, 2.4 at $D = 10$ and 1.43 at $D = \infty$. At large D the field and its gravity weaken: the centre slope of the field falls like $2.72/\sqrt{n}$ and the departure of c from 1 at the centre like $3.65/n$ (Sec. V).

TABLE IV. Held-out test of large-D models (cleaned double-double points; snapshot $D \le 60.1$).

Model	max residual, $10 \le D \le 45$	worst error, $45 < D \le 60.1$	fitted on $D \le 37.5$: worst error	limit
$1/D$, degree 16	4.1×10^{-15}	5.2×10^{-13}	4.2×10^{-11}	1.64389375
$1/D$, degree 18	4.0×10^{-15}	1.8×10^{-12}	1.5×10^{-10}	1.64389377
$1/(D-2)$, degree 18	4.0×10^{-15}	2.8×10^{-12}	7.9×10^{-11}	1.64389377
$(D-2)^{-1/2}$, degree 20	3.9×10^{-15}	2.0×10^{-11}	3.9×10^{-10}	1.636
$D^{-1/2}$, degree 20	3.8×10^{-15}	1.6×10^{-10}	6.2×10^{-9}	1.547

IV. THE LARGE-D LIMIT

A. Is the period analytic in $1/D$?

Each model in Table IV is a polynomial in one variable, fitted at 60 digits on the cleaned double-double points $10 \le D \le 45$ (or ≤ 37.5) and compared with every point above, up to $D = 60.1$, which it never saw (the data snapshot when the table was made; Fig. 3 uses all points to $D = 61.0$). Every model fits its range to the level of the points; the unseen points decide. Fitted on $D \le 45$, the $1/D$ series predicts Δ fifteen units of D ahead to 5×10^{-13}, while the series in half-integer powers miss by 40 to 300 times more, and their limits jump with the degree. Adding one non-analytic term ($D^{-1/2}$, $D^{-3/2}$, $\ln D/D$, $\ln D/D^2$ or $e^{-D/4}$) to the degree-16 series changes neither the prediction nor the limit, and the term's coefficient is not determined.

Fitted on all cleaned points, power series in $1/D$ of degree 16–20 and in $1/(D-2)$ of degree 18–20 fit to the noise and give limits from 1.643893744 to 1.643893752 (snapshot $D \le 60.1$) and from 1.643893745 to 1.643893752 (all 823 cleaned points to $D = 61.03$). Over 48 choices of model ($1/D$, $1/(D-2)$), degree, fit range and cleaning, the limit ranges over 1.643893744–1.643893763 (`hp/dinf_budget.py`); fitting on $D \le 45$ alone pins it only to 2.5×10^{-7}. Hence, if Δ is analytic in $1/D$,

$$\Delta_\infty = 1.64389375 \pm 1.5 \times 10^{-8}, \qquad \omega_\infty = 2\pi/\Delta_\infty = 3.82213589 \pm 4 \times 10^{-8}, \tag{4}$$

and $\Delta(D) \approx \Delta_\infty + 10.37/n + \dots$. The $\pm 1.5 \times 10^{-8}$ is a spread over fit choices, so we tested it on synthetic data with a known limit: the real D grid, noise and junk points like the real ones, and added structures from a distant pole to a complex pole pair and a logarithmic branch point near the data. In all six cases the 48 fits put the known limit inside their range, with the centre 0.8–0.9×10^{-8} above it, so the bar holds and the quoted centre is probably high by about 10^{-8}. The analytic form is the natural default but not proven: as shown in Sec. V, the period is selected by an effect the $1/D$ expansion does not see.

B. The conjecture $\Delta \to 0$

The conjecture $\Delta \to 0$ [17, 18] comes with no rate. Fitted on $10 \le D \le 45$ and compared with $45 < D \le 61$: a power-law approach to 0, D^{-p} times a series in $1/D$ with $p \ge 0.01$, misses the held-out points by 6×10^{-8} or more, against 7×10^{-13} for the analytic series; a series in $1/\ln D$ with zero limit misses them by 1.3×10^{-11} and still gives $\Delta \approx 0.88$ at $D = 10^{30}$; adding a $c/\ln D$ term to the analytic series gives $c \approx 4 \times 10^{-7}$ and leaves the limit at 1.6438937. A power-law approach to 0 is excluded; a logarithmic one is disfavoured but cannot be excluded from $D \le 61$ alone. Bland's hypothesis $\Delta_\infty = 5/2$ [16] is excluded by $\Delta(14) = 2.29$ and decreasing.

C. Candidate closed forms

Five constants that a product search placed inside the window of the earlier double-precision estimate (about nine such coincidences are expected by chance in that search), and four classical constants, were tested. Each was forced as the limit of a power series in $1/D$ fitted on $10 \le D \le 45$ (degree 14 to 26) and compared with the 351 cleaned points from 45 to 59.8 (Table V). All nine are excluded, given analyticity in $1/D$: the closest sits 1.3×10^{-5} from the limit, about 840 times its uncertainty, and every forced fit predicts the unseen points at least 100 times worse than the free fit. Nor is Δ_∞ a product of two of the listed constants with rational prefactor i/j ($i, j \le 12$) and exponents $\pm\frac{1}{2}, \pm 1, \pm 2$ (514 332 expressions; about 0.02 chance hits would be expected in the window). At eight known digits no integer-relation search could support a closed form, so whether Δ_∞ has one is open.

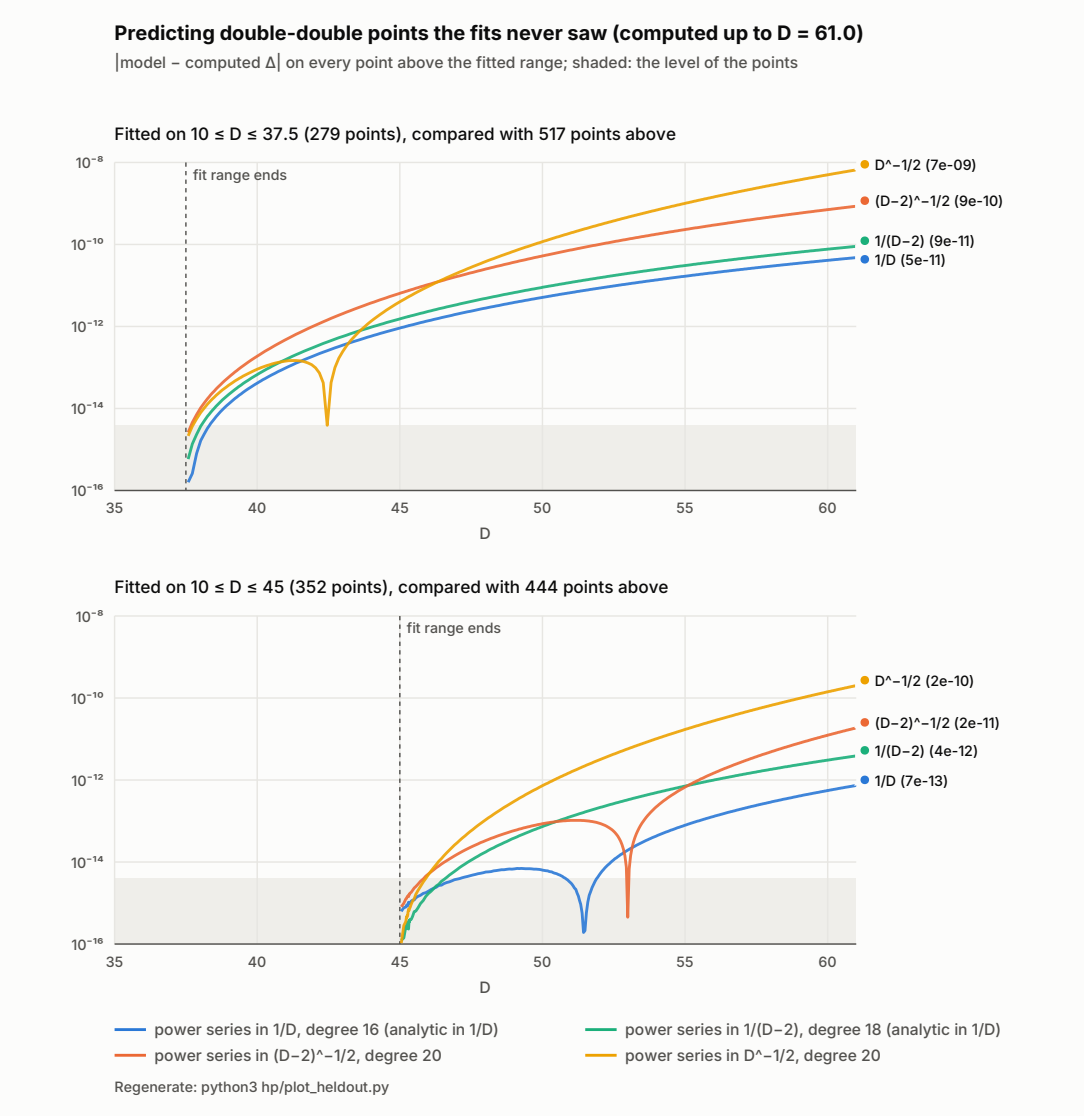


FIG. 3. |model − computed Δ| on every double-double point above the fitted range, for fits on $10 \le D \le 37.5$ (top) and $10 \le D \le 45$ (bottom); shaded: the level of the points.

V. LARGE-D THEORY

A. Scaling

On the double-double solutions at $D = 12$–60: $p \approx xF(\theta)/\sqrt{n}$ near the centre and $p + m$ is smaller by a further factor of order $1/n$; $c = 1 - O(1/n)$ and $b - 1 = (p^2 + m^2)/(n - 1)$ to about 3%; and the centre series has the exact symmetry $m(x) = p(-x)$. The first harmonic of $F = \sqrt{n}\,\partial_x p(0)$ falls from 3.480 at $D = 20$ to 2.962 at $D = 60$ and extrapolates to about 2.72; $n(1 - \langle c(0)\rangle)$ and $\langle F^2\rangle$ approach a common limit near 3.7.

B. The $1/n$ expansion never fixes the period

Write $p = \varepsilon^{1/2}P$, $m = \varepsilon^{1/2}M$, $S = P + M$, $\Delta_P = P - M$, and expand $S = \sum_{k\ge1}\varepsilon^k S_k$, $\Delta_P = \sum_{k\ge0}\varepsilon^k D_k$, $c = 1 - \sum_{k\ge1}\varepsilon^k G_k$, $b = 1 + \sum_{k\ge2}\varepsilon^k B_k$, $\omega = \sum_k \varepsilon^k \omega_k$ (integer powers of ε; this is the scaling measured above, and the statements below hold within this ansatz).

TABLE V. Candidate closed forms for Δ_∞, forced as the limit: best held-out error and its ratio to the free fit. G is Catalan's constant, $j_{\nu,k}$ the k-th zero of J_ν.

Candidate	value	minus Δ_∞	held-out error (ratio)
none (free fit)	$1.64389375 \pm 1.5 \times 10^{-8}$		4.7×10^{-13}
$\frac{5}{7}\sqrt{G}\, j_{0,1}$	1.6439747	$+8.1 \times 10^{-5}$	6.2×10^{-11} (130)
$\frac{7}{5}(\ln 2)^{-2}\pi^{-1/2}$	1.6440014	$+1.1 \times 10^{-4}$	1.1×10^{-10} (220)
$\frac{7}{2}\sqrt{\ln 2/\pi}$	1.6440152	$+1.2 \times 10^{-4}$	1.3×10^{-10} (270)
$\frac{9}{8}\sqrt{j_{2,1}/j_{0,1}}$	1.6440207	$+1.3 \times 10^{-4}$	1.4×10^{-10} (290)
$\frac{10}{9}\sqrt{j_{0,1}/\ln 3}$	1.6439063	$+1.3 \times 10^{-5}$	5.4×10^{-11} (110)
$\pi^2/6$	1.6449341	$+1.0 \times 10^{-3}$	1.7×10^{-9} (3500)
$2\pi/j_{1,1}$	1.6397878	-4.1×10^{-3}	7.0×10^{-9} (15000)
$\sqrt{e}$	1.6487213	$+4.8 \times 10^{-3}$	8.1×10^{-9} (17000)
$(1+\sqrt{5})/2$	1.6180340	-2.6×10^{-2}	4.4×10^{-8} (92000)

Proposition 1 *Let $F(\theta)$ be any smooth 2π-periodic function with odd harmonics only, $\omega_0, \omega_1, \ldots$ any numbers and $K_1(\theta), K_2(\theta), \ldots$ any smooth odd-harmonic functions. Then there is exactly one formal solution with $D_0 = 2xF$ in which, at every order, S_k is an even polynomial in x starting at x^2, $D_k - xK_k$ is an odd polynomial starting at x^3, and G_k, B_{k+1} are even polynomials (B starting at x^2) with $G_k(1) = 0$. Every order is regular at the centre and analytic at the light cone.*

Proof sketch. Induction on k. The b equation gives B_{k+1} algebraically (its coefficient is $(n-1)b \approx 1/\varepsilon$); the c equation gives $-x\partial_x G_k \in x^2 E$ (E: even polynomials with periodic coefficients), so G_k is an even polynomial without $\log x$, fixed by $G_k(1) = 0$; the difference equation gives $S_k \in x^2 E$; and the sum equation reads $x\partial_x D_k - D_k = R_k$ with R_k odd and starting at x^3, because $x\partial_x - 1$ annihilates xK_j. The only possible obstruction, an x^1 term in R_k that would produce $x \log x$, never occurs. Parity in x and the harmonic content in θ are preserved at each step. Polynomials are analytic at $x = 1$, and nothing constrains F or the ω_k. □

The leading-order part of this freedom, any periodic profile with any period, is in [18].

The first orders (sympy, to fourth order) are $P_0 = xF$, $B_2 = 2x^2F^2$, $G_1 = F^2(1-x^2)$, $S_1 = 2x^2(F + \omega_0 F')$ and $D_1 = x^3(\omega_0^2 F'' + 3\omega_0 F' + 2F - 2F^3) + xK_1$. Substituting the expansion through S_3, G_3, B_4 into the exact equations with F from the $D = 60$ solution, the relative residuals fall like n^{-3} for $n = 50$–800, both for $\omega_0 = 2\pi/1.64389$ and for a wrong $\omega_0 = 2\pi/1.4$.

So if $\Delta(D)$ is analytic in $1/D$, its expansion coefficients cannot be computed order by order from the field equations: the selection works through terms smaller than any power of $1/n$. The same holds for the critical exponent: the expansion of growing perturbations $e^{\lambda\tau}$ goes through for any λ (the same induction with $\partial_\tau \to \lambda + \omega\partial_\theta$; checked with sympy through order 2), so $\gamma = 1/\lambda$ is also selected beyond all orders.

Why the ansatz. The scaling is forced by one measured fact, $c - 1 = O(1/n)$ (gravity weak; $n(1 - \langle c(0)\rangle) \to 3.65$): the b equation gives $b - 1 \approx (p^2 + m^2)/(n-1)$, the c equation then gives $c - 1 = O(n(b-1)) = O(p^2)$, so $c - 1 = O(1/n)$ requires $p = O(n^{-1/2})$; and the $O(n)$ terms of the p, m equations force $p + m = O(p/n)$, the measured $p + m = O(n^{-3/2})$. Integer powers of ε beyond the overall $\varepsilon^{1/2}$ are an assumption (analyticity in $1/n$), consistent with the held-out tests on $\Delta(D)$ but not implied by them.

C. The leading-order problem

At leading order ($c = b = 1$) the equations describe a free wave in self-similar form. For one harmonic $e^{i\mu\theta}$ put $s = i\mu\omega$, $\alpha = (1+s)/2$, $\beta = (2+s)/2$, $\kappa = (n+1)/2$.

Proposition 2 *The solution regular at the centre is, up to a constant, $p - m = x\, {}_2F_1(\alpha, \beta; \kappa; x^2)$. Near the light cone it contains the non-analytic branch $(1-x^2)^{n/2-1-s}$ with coefficient*

$$C(n,s) = \frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(s+1-\frac{n}{2}\right)}{\Gamma\left(\frac{1+s}{2}\right)\Gamma\left(\frac{2+s}{2}\right)}, \tag{5}$$

which is finite and non-zero for every real $\omega \neq 0$, odd μ and $n \geq 2$. Hence no time-periodic self-similar solution of the linear problem is analytic at the light cone, for any $D \geq 4$.

Proof sketch. Eliminating $p+m$ gives Gauss's equation for $y = x\,w(x^2)$; regularity (for $n=2$, parity) selects ${}_2F_1(\alpha,\beta;\kappa;z)$, and the connection formula [21] applies because $\kappa-\alpha-\beta = n/2-1-s$ is not an integer; its second coefficient is $C(n,s)$, a ratio of Gamma functions whose arguments have non-zero imaginary parts. □

For the fundamental at $\omega=\omega_\infty$, $|C| = 0.0121$, 0.0114, 0.0124, 0.0192, 0.0405 at $n = 18$, 38, 58, 200, 1000. The free wave equation in similarity coordinates was analysed in 3+1 dimensions by Donninger [20], which contains the $D=4$ case of Proposition 2.

D. What gravity does to the non-analytic piece

The non-analytic piece is a fast WKB mode; in Proposition 2 it is $x^{2-n}(1-x^2)^{n/2-1-s}$.

Proposition 3 *Let $c = 1-\varepsilon F^2(1-x^2)+O(\varepsilon^2)$ as in Proposition 1, and $\Phi(x) = \ln x - \frac12\ln(1-x^2)$. Perturbations $\delta p = e^{-n\Phi}(A+O(1/n))$, $\delta m = e^{-n\Phi}\big(\frac{1-x}{1+x}A+O(1/n)\big)$ solve the linearized equations on $0<x<1$ to this order if and only if*

$$x(1-x^2)\,\partial_x A - 2\omega x^2\,\partial_\theta A = (1+x-2x^2F^2)\,A, \tag{6}$$

i.e. along the characteristics $\theta = \theta_0+\omega\ln(1-x^2)$, $A = A_0(\theta_0)\,\frac{x}{1-x}\exp\big(-\int_0^x \frac{2tF(\theta_0+\omega\ln(1-t^2))^2}{1-t^2}\,dt\big)$.

Proof sketch. The order-n terms give $x(1-x^2)\Phi' = 1$ and $\delta m/\delta p = (1-x)/(1+x)$; at order 1 the corrections satisfy a singular linear system whose left null vector $(1+x,-(1-x))$ gives the transport equation as the solvability condition. Perturbations of b, c and $G\,\delta p$ enter one order later (confirmed symbolically from the exact equations). □

Integrating the linearized equations directly from $x=0.9$ to 0.72 ($F = 2.72\sin\theta + 0.22\sin 3\theta$, $\omega=\omega_\infty$) reproduces the transported A with relative errors 0.57, 0.26, 0.13, 0.062, 0.031 at $n = 100$–1600, i.e. like $1/n$; with the full $O(1/n)$ background the numbers are nearly the same. Three consequences: (i) gravity acts on the non-analytic piece at order 1 (over this interval it multiplies it by about 40), and shifts the light-cone exponent by $\langle F^2\rangle \approx 3.7$, so an expansion in the wave amplitude is not a route to the selection; (ii) the factor never vanishes, so gravity cannot cancel the piece on the way to the light cone; (iii) every characteristic reaches $x=0$, where the fast mode is the singular branch x^{2-n}, so a regular solution carries no fast mode on $0<x<1$ at any order of this WKB expansion. The non-analytic piece must come from a Stokes phenomenon.

E. The light-cone coefficient as a Stokes multiplier

A divergent series records the exponentially small terms it switches on in its late orders [22]. At leading order this can be made exact, because the $1/n$ expansion of $x\,{}_2F_1$ from Proposition 2 is explicit.

Proposition 4 *Let $0<x<1$, $\Phi = \ln x - \frac12\ln(1-x^2)$, $\tau_\ell = \Phi - i\pi(\ell+\frac12)$, and let $y = x\,{}_2F_1(\alpha,\beta;\kappa;x^2)$ with $\kappa = (n+1)/2$, $y\sim\sum_k y_k\varepsilon^k$, $\varepsilon = 1/n$. If $|\Phi| < (\sqrt3/2)\pi$ (equivalently $0.066 < x < 0.998$), then*

$$y_k = K\,x^2(1-x^2)^{-1-s}\,\frac{\Gamma(k+s+\frac12)}{\tau_0^{\,k+s+\frac12}}\,\big(1+O(1/k)\big), \qquad K = e^{-i\pi(1+s)}\,\frac{2^{-s-1/2}}{\Gamma(\alpha)\Gamma(\beta)}. \tag{7}$$

The $\ell=-1$ singularity contributes at the same rate but with a k-independent relative weight of 8×10^{-5} ($x=0.5$) to 5×10^{-8} ($x=0.95$). If $|\Phi| > (\sqrt3/2)\pi$ the late orders are governed instead by the x-independent singularities $\pm i\pi$.

Proof (condensed). (i) *Euler integral.* For $n>1$, ${}_2F_1 = \frac{\Gamma(\kappa)}{\Gamma(\beta)\Gamma(\kappa-\beta)}\int_0^1 t^{\beta-1}(1-t)^{\kappa-\beta-1}(1-zt)^{-\alpha}dt$; with $1-t=e^{-2\tau}$, $y = x\,P(\kappa)\cdot 2\int_0^\infty e^{-\tau/\varepsilon}G(\tau)\,d\tau$, where $G = e^{(2\beta-1)\tau}(1-e^{-2\tau})^{\beta-1}(1-z+ze^{-2\tau})^{-\alpha}$ and $P = \Gamma(\kappa)/(\Gamma(\beta)\Gamma(\kappa-\beta))$. (ii) *Watson's lemma.* Write $G=\tau^{\beta-1}H$; H is analytic except at the simple zeros $\tau=\tau_\ell$ of $1-z+ze^{-2\tau}$, where $H\propto(1-\tau/\tau_\ell)^{-\alpha}$, and at the branch points $\tau = i\pi m$, $m\neq0$. On $\tau>0$, $|G|\le Ce^{\tau}$, so $2\int_0^\infty e^{-\tau/\varepsilon}G\sim 2\sum_m H_m\Gamma(m+\beta)\varepsilon^{m+\beta}$. (iii) *Darboux.* The radius of convergence of $\sum H_m\tau^m$ is $\min(|\tau_0|,\pi)$; if $|\tau_0|<\pi$ the only singularities on that circle are τ_0 and τ_{-1}, giving $H_m = \sum_{\ell=0,-1} h_\ell\,\frac{\Gamma(m+\alpha)}{\Gamma(\alpha)\,m!}\,\tau_\ell^{-m}(1+O(1/m))$. (iv) *Prefactor.* $P(\kappa)(2\varepsilon)^\beta(1+\varepsilon)^{-\beta}\Gamma(\beta) = 1+\sum_j p_j\varepsilon^j$, from Binet's expansion of $\ln\Gamma(\kappa)-\ln\Gamma(\kappa-\beta)$, whose Borel transform is singular only at $\tau=i\pi m$, a larger radius. (v) *Product lemma.* If $A=\sum a_k\varepsilon^k$ has Borel radius ρ_A and $B=\sum b_j\varepsilon^j$ has $\rho_B>\rho_A$, then $(AB)_k = b_0a_k(1+O(1/k))$; hence $y_k = \frac{x\,2^{1-\beta}}{\Gamma(\beta)}\Gamma(k+\beta)H_k(1+O(1/k))$. (vi) *Constant.* $\Gamma(k+\alpha)\Gamma(k+\beta)/\Gamma(k+1) = \Gamma(k+s+\frac12)(1+O(1/k))$,

and evaluating h_0 in closed form (with $e^{2\tau_0} = z/(z-1)$) gives the prefactor $x^2(1-x^2)^{-1-s}$ times K. If $|\tau_0| > \pi$ the radius is π and step (iii) applies to $\pm i\pi$ instead. □

Numerically, the late orders computed to order 122 at 90 digits agree: at $x = 0.9$, y_k/prediction -1 extrapolates to $(-0.9 - 2.3i) \times 10^{-3}$ over $k = 100$–122; the derived constant equals K to $1 + 3.65/k$ at $x = 0.5$, 0.8 and 0.9; and at $x = 0.999$ and 0.9999 the measured singulant is $-3.135i \to -i\pi$, not $\Phi - i\pi/2$, as the statement predicts. The singularities $\pm i\pi$ come from the Gamma-function prefactor and the endpoint factor $(1 - e^{-2\tau})^{\beta-1}$; their exponentials $e^{\mp i\pi n}$ are pure phases, not exponentially small terms, so they do not affect the Stokes-line argument, which concerns the τ_ℓ.

Hence, with $\chi = \tau_0 = \Phi - i\pi/2$: $e^{-n\chi} = e^{in\pi/2}x^{-n}(1-x^2)^{n/2}$ is the fast mode of Proposition 3; χ vanishes only at $x = \infty$, so the mode is born at the outer singular point; its Stokes line is the real axis $x > 1$ approached from above, where $\chi = \ln(x/\sqrt{x^2-1})$; on $0 < x < 1$ nothing is switched on (the optimally truncated series matches the exact solution to 4×10^{-40} at $x = 0.9$, $n = 60$, against a singular-branch size of 7×10^{-21}). The Stokes line ends at the light cone, and $C(n, s)$ is the Stokes multiplier.

F. The inner problem

At $x \sim \sqrt{n}$ the expansion is not uniform: gravity is of order one and fast and slow modes are not separated. In physical variables this is the region $r = O(\sqrt{n})$ at fixed t, where the field equations at leading order are, up to a change of variables, the large-D system of Rozali and Way [8] (see Proposition 5). With $\xi = x/\sqrt{n}$, $y = p - m$, $\sigma = \sqrt{n}\,(p + m)$, $b - 1 = \eta/n$ and $L = \xi^2\partial_\xi + \omega\xi\partial_\theta$, the equations become, at leading order,

$$L\,y = c\,\sigma, \qquad L\,\sigma = c\,\xi\,\partial_\xi y + c\,(\tfrac{1}{2}y^2 - 1)\,y, \qquad \xi\,\partial_\xi c = \tfrac{1}{2}\,c\,y^2, \qquad \eta = \tfrac{1}{2}y^2, \tag{8}$$

with no large parameter left. Its formal solution regular at $\xi = 0$ reproduces the top-degree coefficients of the outer polynomials of Proposition 1 (checked to 2×10^{-13} at orders 1–10). Without gravity, with $t = 1/\xi$ and $w = t^{-s}y$ along characteristics, it reduces to Hermite's (Weber's) equation; the regular solution is $y = 2\xi\,{}_2F_0(\alpha, \beta; ; 2\xi^2)$, whose Stokes constant gives, by the duplication formula,

$$C(n, s) \approx -\,i\,\frac{\sqrt{2\pi}}{\Gamma(1+s)}\,(n+1)^{s+1/2}\,e^{-i\pi\Lambda}\,\big(1 + O(s^2/n)\big), \qquad \Lambda = \tfrac{n}{2} - 1 - s, \tag{9}$$

Weber's classical Stokes constant [23]; the ratio to the exact (5) is 1.00014 at $n = 10^5$ ($\mu = 1$).

In the inner region the fast mode obeys $\xi\partial_\xi a + 2\omega\partial_\theta a = 2F^2 a$ (the large-x limit of Proposition 3), so the fast-type Stokes multiplier is a function of the characteristic label, $S(\theta_0) = \sum_\nu S_\nu e^{i\nu\theta_0}$ with $\theta_0 = \theta - 2\omega\ln\xi$; the Borel singularity stays at $\zeta = \frac{1}{2}$ and the exponents shift by $\langle F^2\rangle$. The selection rule is stated as a conjecture.

Conjecture 1 (selection at $D = \infty$) *For $n = D - 2$ let $F_n(\theta) = \sqrt{n}\,\partial_x p(0, \theta)$ and $\omega_n = 2\pi/\Delta(D)$ be the profile and frequency of the critical solution. Assume (H1) $F_n \to F_\infty$ uniformly on a strip $|\mathrm{Im}\,\theta| < \sigma$ for some $\sigma > 0$, and $\omega_n \to \omega_\infty > 0$ (the data give $\sigma \approx 1.4$, the harmonic decay rate); and (H2) the formal family of Proposition 1 with data (F_n, ω_n, K_k) is the asymptotic expansion of the solution uniformly on $\delta \le x \le 1 - \delta$ (checked: residuals $\propto n^{-3}$ through third order). Then $S_\nu(F_\infty, \omega_\infty) = 0$ for every odd ν. Conversely, a zero of S at which the Jacobian with respect to (F, ω), modulo the phase, is invertible should continue to critical solutions for all large n.*

Argument. The inner problem resums the outer series' top-degree terms; the outer series' light-cone exponential is fixed by the inner singularity at $x = \infty$ (exact in the linear case, where the outer late-order constant of Proposition 4 is the inner Weber Stokes constant); and analyticity at the light cone forces $C = 0$. The map $S \mapsto C$ is, at leading order in $1/n$, a composition of three invertible maps: (1) inner–outer matching on $1 \ll x \ll \sqrt{n}$, where the inner fast mode $e^{-1/(2\xi^2)}\xi^{2\langle F^2\rangle}e^{\Pi(\theta)}S(\theta_0)$, with $\Pi = \omega^{-1}\int(F^2 - \langle F^2\rangle)$, matches the outer WKB mode $e^{-n\Phi}A$ on the branch above the real axis $x > 1$; (2) the outer transport of Proposition 3 along $x > 1$ with real θ, which has real coefficients while $\varepsilon x^2 F^2 \ll 1$ and acts along real characteristics as multiplication by a non-vanishing function composed with a shift; and (3) the half-monodromy at the regular singular point $x = 1$, which continues the singular Floquet branch $(1-x)^\Lambda\varphi(\theta - \omega'\ln(1-x))$, $\omega' = \omega/(1 - \langle G(1)\rangle)$, from $x > 1$ to $x < 1$ and multiplies harmonic μ by $e^{-i\pi\Lambda}e^{\pm\pi\mu\omega'}$. Hence $C \equiv 0 \Leftrightarrow S \equiv 0$ at leading order; in the linear case this is exact.

Rigour fails at three points (a staged proof roadmap, with the selection theorem for an M-harmonic model written out lemma by lemma, is in the repository's PROOF.md). (a) Borel summability of the nonlinear series is not proved; the outer nonlinear late orders also carry x-independent singularities of modulus about 0.13, driven by high harmonics, and one must show that these give only ambiguities in the free data, not light-cone branches (in the linear case the analogous $\pm i\pi$ are harmless pure phases, which is suggestive but not a proof). (b) Uniformity in the harmonic index

(narrower than it looks: stated without harmonics the condition uses F only on the real line, and the weights below cancel in every tuning equation, so a tail lemma is needed only for the jump-conditioned truncation, whose tail beyond $M = 7$ is estimated at a tenth of what a 10^{-3} certificate tolerates): the weights $e^{\pi|\mu|\omega/2}$ grow like $e^{6.0|\mu|}$ while F's harmonics decay only like $e^{-1.4|\mu|}$, so an $O(1/n)$ error in a high harmonic is amplified when it mixes into low harmonics through e^{Π}; a proof needs a function-space formulation, e.g. on the strip $|\mathrm{Im}\,\theta| < \sigma$ with weights in a dual norm. (c) Uniformity at $x = 1$: the outer WKB amplitude is singular there ($A \propto (1-x)^{-1}$) and must be matched to the Frobenius solution on $1/n \ll |1-x| \ll 1$; the exponents agree ($\Lambda = n/2 - 1 - s + \langle F^2\rangle + O(1/n)$ on both sides), but the matching has not been written out.

Computations with gravity. Two independent implementations (lateral solutions on either side of the Stokes line, and Borel–Darboux fits in ball arithmetic) reproduce the linear constant to 10^{-11} and 6×10^{-27}. With $F = A\sin\theta$ alone, both find an isolated zero of the multiplier at $A = 2.542148$, $\omega = 2.969777$ (a computer-assisted proof in ball arithmetic, following the template of Baldomá et al. [6] for inner Stokes constants, with a validated Lohner integrator and Krawczyk's test, certifies that one component of the jump at one radius has exactly one zero, a simple one, within 10^{-9} of $(A, \omega/\omega_\infty) = (2.5421480475, 0.7769940136)$; that this component vanishing forces the whole jump to vanish is proved, and so is the analyticity near $\xi = 0$ that identifies the computed solution with the Borel sum, which leaves one lemma, analyticity on an annulus $0.0624 \le |\xi| \le 0.2$, for the full one-harmonic condition; an independent C++ implementation with the CAPD library confirms it and certifies a box ten times larger; repository, proof/A_certify.md and A_review.md; Reiterer and Trubowitz [5] gave a computer-assisted existence proof of the $D = 4$ critical solution itself); it is not the physical solution (ω 22% low) and does not survive adding harmonic 3. Truncating harmonics is not a controlled approximation here: the linear multipliers grow like $F_\nu e^{\pi|\nu|\omega/2}$.

Three harmonics. With $F = A\sin\theta + (F_3 e^{3i\theta} + \text{c.c.})$ the joint solve of $S_1 = S_3 = 0$ for (A, ω, F_3) is badly conditioned (Jacobian singular values 6×10^9, 5×10^9, 23, 0.4): the de-mixed harmonic-1 residual picks up about 10^4 times the harmonic-3 one ($\sim e^{\pi\omega}$ times the finite-ξ de-mixing error), and the joint solve stops on false near-zeros where a small S_3 cancels S_1. A nested solve removes this: at fixed (A, ω) solve $S_3 = 0$ for F_3 to 10^{-11}, then apply Newton's method to (A, ω) for $S_1 = 0$ (a 2×2 problem of condition about 2). It has one zero, the same from two starts and at three matching radii (moving by less than 10^{-7}): $A = 2.7265$, $\omega = 1.0156\,\omega_\infty$ ($\Delta = 1.6186$), $F_3/F_1 = 0.0530 + 0.0041i$, $\langle F^2\rangle = 3.727$. The shape is that of the $D = \infty$ solution extrapolated from the data ($A \approx 2.70$–2.72, $|F_3/F_1| \approx 0.050$–0.058, $\langle F^2\rangle \approx 3.65$–$3.7$), and ω is 1.56% above ω_∞, against 22% below with one harmonic.

Five harmonics. At $M = 5$ the couplings grow further ($\partial r_3/\partial B_5 \approx 10^{13}$, $\partial r_1/\partial B_5 \approx 10^{17}$) and the nested solve's homotopy stalls. A hierarchical solve converges: $S_5 = 0$ innermost, then $S_3 = 0$, then $S_1 = 0$ for (A, ω), each level a damped 2×2 Newton step judged by step size, with S_5 first solved at a small radius where its jump is linear and then polished at the matching radius. At fixed $(2.72, \omega_\infty)$ it gives $F_3/F_1 = 0.05486 + 0.00541i$ and $F_5/F_1 = 0.00309 + 0.00054i$, against the amplitude expansion's $0.05482 + 0.00543i$ and $0.00310 + 0.00056i$ (two methods sharing no code; the differences are of the size of the omitted F_7). The full selection, iterated to its noise floor, gives $A = 2.71858$, $\omega = 0.99923\,\omega_\infty$ ($\Delta = 1.64516$, 0.077% above Δ_∞; steady to 8×10^{-6} in ω/ω_∞ over the last steps). The offset in ω thus goes -22%, $+1.56\%$, -0.077% for $M = 1$, 3, 5, alternating in sign and shrinking fifteen- to twentyfold per added pair of harmonics; Aitken's extrapolation of the three gives $\omega/\omega_\infty = 1.0003$.

Amplitude expansion. Writing $F = A\sin\theta + \sum_{\nu\ge 3} f_\nu(A) e^{i\nu\theta}$ (plus c.c.), harmonic μ first appears at order $A^{|\mu|}$, so every harmonic is kept exactly. Tuning f_ν order by order so that the multipliers of all harmonics $|\nu| \ge 3$ vanish, at $A = 2.72$ and $\omega = \omega_\infty$, gives $|F_3/F_1| = 0.0551$ and $|F_5/F_1| = 0.00314$, with ratios $f_{\nu+2}/f_\nu \approx qA^2$, $|q|A^2 \approx 0.063$, nearly independent of ν. The double-double solutions at $D = 20$–60, extrapolated in $1/n$ (degree 2–3), give $|F_3/F_1| \to 0.0552$–0.0549 and $|F_5/F_1| \to 0.0034$–0.0032, and the ratios $|F_{\nu+2}/F_\nu|$ ($\nu = 1, \ldots, 9$) extrapolate to 0.055–0.072, a finite decay rate of 1.3–1.45 per harmonic, as predicted (Fig. 5; the extrapolations are not fully converged). This is the first number the inner problem has produced that can be checked against the solutions, and it agrees. The same relation evaluated at each D's own amplitude and ω (a heuristic) tracks the measured F_3/F_1 at $D = 20$–60 to within 0.004–0.006 (0.126, 0.099, 0.087, 0.080, 0.075 against 0.130, 0.105, 0.093, 0.085, 0.080).

The harmonic-1 condition. With the higher harmonics tuned, $S_1/S_1^{(1)} = 1 + \sigma_1 A^2 + \sigma_2 A^4 + \ldots$. The late orders of harmonic ν at order k contain every neighbour family ν' with $|\nu'| \le k$ (they enter through the label θ_0 at relative order $1/j$), and the fits must include them all; a basis with only $\nu' - \nu = \pm 2, \pm 4, \pm 6$ is good through σ_4 but fails from σ_6 on. With all families, in ball arithmetic at 640 bits, the values at ω_∞ agree between $J = 750$ and 1000 terms to about 10^{-19}: $\sigma_1 = 0.74358457992 + 0.52447485276i$, $\sigma_2 = 0.11879383784 + 0.39807457035i$, $\sigma_3 = -0.05397185243 + 0.11658742432i$, $\sigma_4 = -0.02886475753 + 0.01077835879i$, $\sigma_5 = -0.00555413676 - 0.00280595708i$, $\sigma_6 = (-2.7853745790 - 10.776628405i) \times 10^{-4}$, $\sigma_7 = (1.0223846436 - 1.4905053530i) \times 10^{-4}$.

At ω_∞ the coefficients τ_k of $\ln S_1$ in $x = A^2$ have $\arg\tau_k/\pi = 0.878$, 0.944, 0.978, 0.986, 0.992, 0.995 for $k = 2$–7, approaching π, as a real zero x_0 requires ($\tau_k \to -1/(kx_0^k)$; a complex x_0 would turn them with k). Padé approximants [1/5] to [5/1] of $(\ln S_1)'$ put the zero at $A_0 = 2.728$–2.756 with $\mathrm{Im}\,A_0 = -0.013$ to -0.027 (residues 1.05–1.22), and ratio estimates from the last coefficients give $2.744 - 0.026i$ and $2.748 - 0.013i$. The amplitude extrapolated from the solutions is 2.70–2.72, so at ω_∞ the harmonic-1 condition has a nearly real zero within about 1% of it.

Across ω two families of estimates of where the zero is exactly real approach ω_∞ from opposite sides (Table VI, Fig. 4; σ_1–σ_{12} at $\omega/\omega_\infty = 0.99$, 1.00, 1.01, σ_1–σ_8 at five values 0.98–1.02, σ_1–σ_7 at seven values 0.97–1.03). The crossings $\arg\tau_k = \pi$ assume a simple zero, $\tau_k \to -1/(kx_0^k)$. The coefficients behave as $-c/(kx_0^k)$ with an effective exponent c that falls toward 1 as orders are added (last column), so x_0 is a simple zero, the departure of c from 1 is contamination by singularities beyond it, and the crossings from above are the asymptotically correct estimator. Their distance from the limit shrinks by a factor of about 0.4–0.6 per order, to 1.00011 at $k = 12$ and still falling, and Aitken's extrapolations of consecutive triples scatter around 1 at the 10^{-4} level. The ratio estimates, which let c float, come up from below more slowly; Padé approximants of $(\ln S_1)'$ put $\operatorname{Im} A_0 = 0$ at 0.981–0.991 with seven coefficients, 0.987–0.998 with eight, 0.992–0.998 with nine, and with ten to twelve they straddle ω_∞ (with twelve, the five central approximants agree at 0.99987–0.99991, and at ω_∞ they are real to $|\operatorname{Im} A_0| \approx 1.4 \times 10^{-4}$, at $A_0 = 2.7196$). At twelve orders the bracket is 0.99915–1.00011. A third estimator agrees: the ratios σ_k/σ_{k+1} grow linearly in k, so S_1 is numerically an entire function of A^2 of exponential type, and the series of $e^{-\bar\kappa A^2} S_1$ (same zeros, no cancellation) puts the real zero at $\omega/\omega_\infty = 0.99988$ at twelve orders, with $A_0 = 2.7194$ isolated from the other zeros. The five-harmonic truncation (0.999228) lies inside the bracket. The selection rule thus picks $\omega/\omega_\infty = 0.9999$–$1.0001$ at amplitude $A_0 = 2.720$ (the solutions: 2.70–2.72): $\Delta_\infty = 1.6439 \pm 0.0002$ from the $D = \infty$ inner equations alone, with no input from the finite-D solutions, against 1.64389375 from them. Whether the remaining 10^{-4} (the Padé value lies 1.1×10^{-4} below 1) closes needs more orders.

TABLE VI. Where the zero of S_1 becomes real, in ω/ω_∞, order by order.

order k	$\arg\tau_k = \pi$ (from above)	ratio τ_{k-1}/τ_k real (from below)	effective exponent c at ω_∞
5	1.0083	0.9821	
6	1.0040	0.9803	$1.148 - 0.159i$
7	1.0021	0.9913	$1.177 - 0.095i$
8	1.0011	0.9946	$1.077 - 0.059i$
9	1.0006	0.9961	$1.052 - 0.044i$
10	1.0003	0.9978	$1.021 - 0.026i$
11	1.00020	0.99919	$1.017 - 0.012i$
12	1.00011	0.99915	$1.013 - 0.012i$

The critical exponent. The double-double data give $\lambda_\infty = 2$ to about 10^{-6} (Sec. III), and the inner problem explains it. Proposition 1 holds for perturbations as well, so λ, like ω, is selected beyond all orders; but one growth rate of the inner problem is exact. In physical inner variables, $\rho = r/\sqrt{n}$ and t, one has $\xi = \rho/(-t)$, $L = \rho\,\partial_t$ at fixed ρ and $\xi\partial_\xi = \rho\partial_\rho$ at fixed t. Put $s = \rho^2/2 = r^2/(2n)$, $y = \sqrt{s}\,Y$ and $\sigma = \sqrt{2}\,s\,S$.

Proposition 5 (an exact unstable mode at $D = \infty$) *(i) In these variables the inner equations are the nonlinear heat equation (Rozali and Way's leading-order system, below)*

$$Y_t = cS, \qquad S_t = c\left(Y_s + \tfrac{1}{4}Y^3\right), \qquad c_s = \tfrac{1}{4}cY^2, \tag{10}$$

with no explicit s or t; they are invariant under s-translation, t-translation, the scaling $(t, s, Y, S) \to (at, a^2 s, Y/a, S/a^2)$ and the reparametrizations $t \to T(t)$, $c \to T'c$. (ii) Let $(y, \sigma, c)(\xi, \theta)$ be a DSS inner solution with frequency ω, regular at $\xi = 0$, with $y = 2F\xi + O(\xi^3)$ and $c = 1 + F^2\xi^2 + O(\xi^4)$. Let P be the periodic solution of $P + \omega P' = -2F^2$ and $X = \xi\partial_\xi + \omega\partial_\theta$. Then

$$\delta y = e^{2\tau}\Big(\frac{\xi y_\xi - y}{\xi^2} + P\,Xy\Big), \quad \delta\sigma = e^{2\tau}\Big(\frac{\xi\sigma_\xi - 2\sigma}{\xi^2} + P\,X\sigma\Big), \quad \delta c = e^{2\tau}\Big(\frac{c_\xi}{\xi} + (P + \omega P')\,c + P\,Xc\Big) \tag{11}$$

solves the linearized inner equations: an exact perturbation with $\lambda = 2$. It is regular at $\xi = 0$, keeps $c(t, 0) = 1$, and has leading profile $\delta y = 2f\xi e^{2\tau} + O(\xi^3)$, $f = (2 + \omega\partial_\theta)(F + \omega F') - 2F^3 + P\,(F + \omega F')$. (iii) Wherever the fast-type Stokes multipliers of the background are defined by Borel summation, those of this perturbation are a linear image of them and vanish with them; so under Conjecture 1, $\lambda = 2$ is an exact growth rate of the $D = \infty$ critical solution.

Proof sketch. (i) Direct substitution (sympy, residual 0); no coefficient depends on s or t. This system is Rozali and Way's leading-order flat-space equations [8]: with their r, $\tau = \sqrt{D}\,t$ and lapse factor $1 - \delta_0$ identified with our ρ, t and c^2, their equations become $\phi_s = c^{-1}\partial_t(c^{-1}\phi_t)$ and $c_s = \phi_t^2/c$, and $Y = 2\phi_t/c$, $S = Y_t/c$ then satisfy the three equations above identically (sympy). They noted its parabolic character with space and time swapped; what is new here is the symmetry argument for the DSS solution. (ii) With $\tau = -\ln(-t)$, $\partial_t = e^\tau X$ on a DSS background and $\partial_s = e^{2\tau}\xi^{-2}\,\xi\partial_\xi$ at fixed t, so a generator of scaling weight k gives a perturbation $\propto e^{k\tau}$; s-translation has

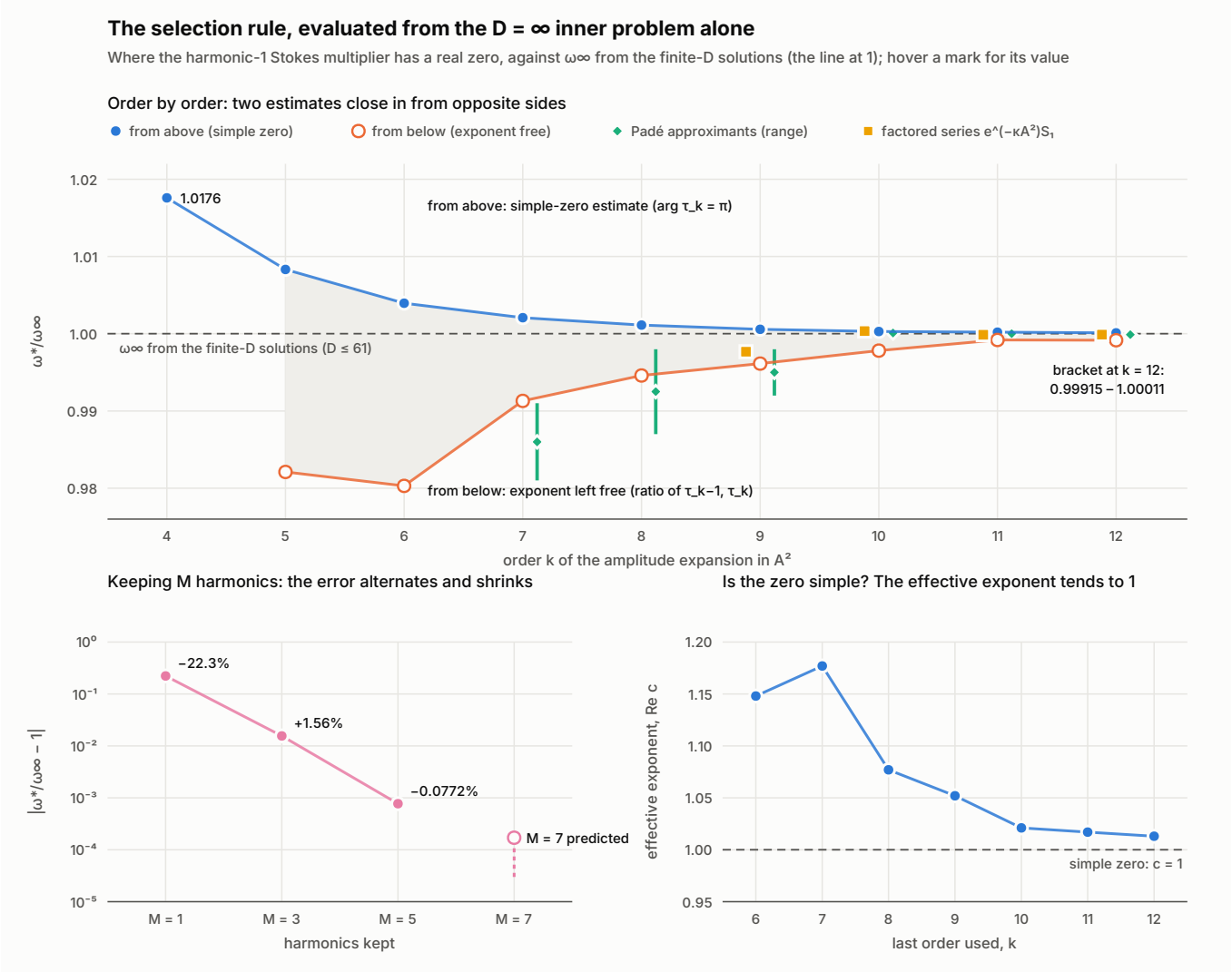


FIG. 4. The selection rule evaluated from the $D = \infty$ inner problem alone. Top: where the harmonic-1 multiplier has a real zero, order by order in the amplitude expansion; simple-zero estimates ($\arg \tau_k = \pi$) from above, floating-exponent estimates from below, Padé ranges and the factored series inside the shaded bracket, which narrows onto ω_∞ from the finite-D solutions (dashed). Bottom left: the truncated route, $|\omega^*/\omega_\infty - 1|$ for $M = 1$, 3, 5 harmonics (alternating in sign) and the $M = 7$ prediction. Bottom right: the effective exponent of the zero tends to 1.

weight 2. It changes c at the centre by $2F^2e^{2\tau}$, which the reparametrization $t \to t - e^\tau P(\theta)$ cancels exactly when $P + \omega P' = -2F^2$, solvable harmonic by harmonic since $1 + i\nu\omega \neq 0$. With $y = \sum_j d_j \xi^{2j+1}$, the s-translation part is $\sum_k 2(k+1)d_{k+1}\xi^{2k+1}$, and X preserves parity. (iii) $\delta Z = \mathcal{D}Z$ for a first-order operator $\mathcal{D}$ with coefficients analytic away from $\xi = 0$; the regular formal solution with given leading profile is unique, and lateral Borel sums commute with $\mathcal{D}$, so the jump of δZ is $\mathcal{D}$ applied to the background's jump. □

As a check, the perturbation recursion (gravity included, $\delta c(0) = 0$ imposed, nothing about symmetries built in) run at $\lambda = 2$ from f reproduces the predicted coefficients to 10^{-180} in every resolved harmonic, while $\lambda = 1.9$ or 2.1 miss by 7.5% at the third order. The mode is physical, not gauge: the reparametrization is gauge and t-translation ($\lambda = 1$) shifts the accumulation time, but s-translation shifts r^2, with r the areal radius. So $r_{\rm BH}^2 \propto p - p^*$ and $\gamma_\infty = 1/2$, the DSS counterpart of Rozali and Way's estimate [8], exact at $D = \infty$ given the conditions stated next: *if Conjecture 1 holds, the $D = \infty$ solution has no unstable mode other than $\lambda = 2$ and the gauge mode $\lambda = 1$, and $\lambda(D)$ is continuous at $1/n = 0$, then $\gamma_\infty = 1/2$.* At finite D the outer region and the $1/n$ corrections depend on r explicitly, so the mode survives as an ordinary unstable mode with $\lambda = 2 + O(1/n)$; the data give $2 + 3.0324/n$. Open are the uniqueness of the unstable mode at $D = \infty$ and the coefficient 3.0324, which needs the $O(1/n)$ inner equations.

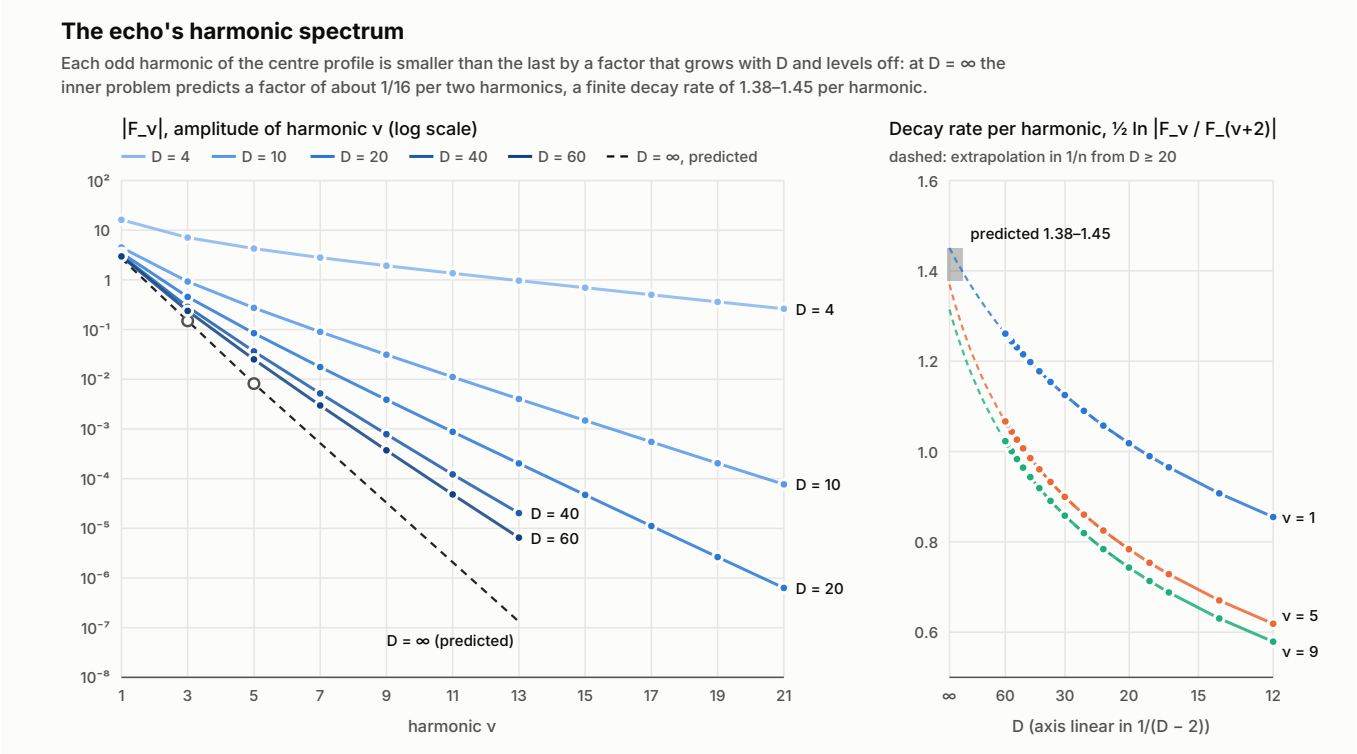


FIG. 5. Left: amplitude $|F_\nu|$ of each odd harmonic of the echo profile for $D = 4$–60 and the $D = \infty$ prediction of the inner problem (dashed; beyond $\nu = 5$ the leading-order ratio, conjectured). Right: the decay rate per harmonic, $\frac{1}{2}\ln|F_\nu/F_{\nu+2}|$, against $1/(D-2)$, with extrapolations from $D \geq 20$ and the predicted $D = \infty$ band.

G. Relation to the next-to-leading-order condition of Ref. [18]

Ecker, Ecker and Grumiller use the same gauge: their $f = \alpha/a$ is our c, with $f = 1$ at the self-similarity horizon $x = 1$, and their profile is $\beta(\tau) \approx \sqrt{2/n}\,F(\omega\tau)$. At next-to-leading order they require $f \leq 1$ near the horizon at the zeros of β, which gives their Eq. (20); in our variables, $\omega^2F'' + 3\omega F' = 0$ at the zeros of F. Within our framework this condition does not select Δ, for the following reasons. The inequality $f \leq 1$ holds for every regular solution, whatever the period: from $xb_x = b(p^2+m^2) - (n-1)b(b-1)$ with $b(0) = 1$, b can never drop below 1, so $xc_x = (n-1)c(b-1) \geq 0$ and $c \leq c(1) = 1$. In the formal family of Proposition 1, near a zero of F,

$$b - 1 = 2\varepsilon^2x^2F(\theta - \delta\theta)^2 + 2\varepsilon^4\omega^2x^4F'^2 + \ldots, \qquad \delta\theta(x) = -\varepsilon\omega x^2\,\frac{3F' + \omega F''}{2F'}, \tag{12}$$

so the next-to-leading-order term is the cross term of a square whose square term arrives one order later; the discriminant $16\omega^2x^6F'^4$ is positive for every F and ω. The double-double solutions are consistent with this: they violate Eq. (20) [$(\omega^2F'' + 3\omega F')/(3\omega F') = 0.98$–1.06 at $D = 12$–60, since F is nearly antisymmetric about its zeros], the zero line of $p - m$ tilts as $\delta\theta(x)$ predicts (measured over predicted at $x = 1$: 0.69 to 0.88 for $D = 20$–60), and the maximum of c at the centre approaches $1 - (\varepsilon^3/2)\omega^2F'^2$, their Eq. (22) in this scaling (ratio 0.36 to 0.86). The same next-to-leading-order condition is used by Lai and Wen [4], who take it to select the period; the argument above applies to that use as well.

VI. DISCUSSION

The numerics give $\Delta(D)$ to high precision up to $D = 61$, and they favour a finite limit $\Delta_\infty = 1.64389375$ strongly, under the assumption of analyticity in $1/D$ that they support but cannot prove. The theory explains why that number cannot come from the $1/D$ expansion: the expansion is unobstructed for every period, and the selection acts through an exponentially small, Stokes-type effect whose linear version is solved exactly. The inner problem at $x \sim \sqrt{n}$ is n-free; its higher-harmonic conditions already reproduce the shape of the echo at $D = \infty$, and its harmonic-1 condition

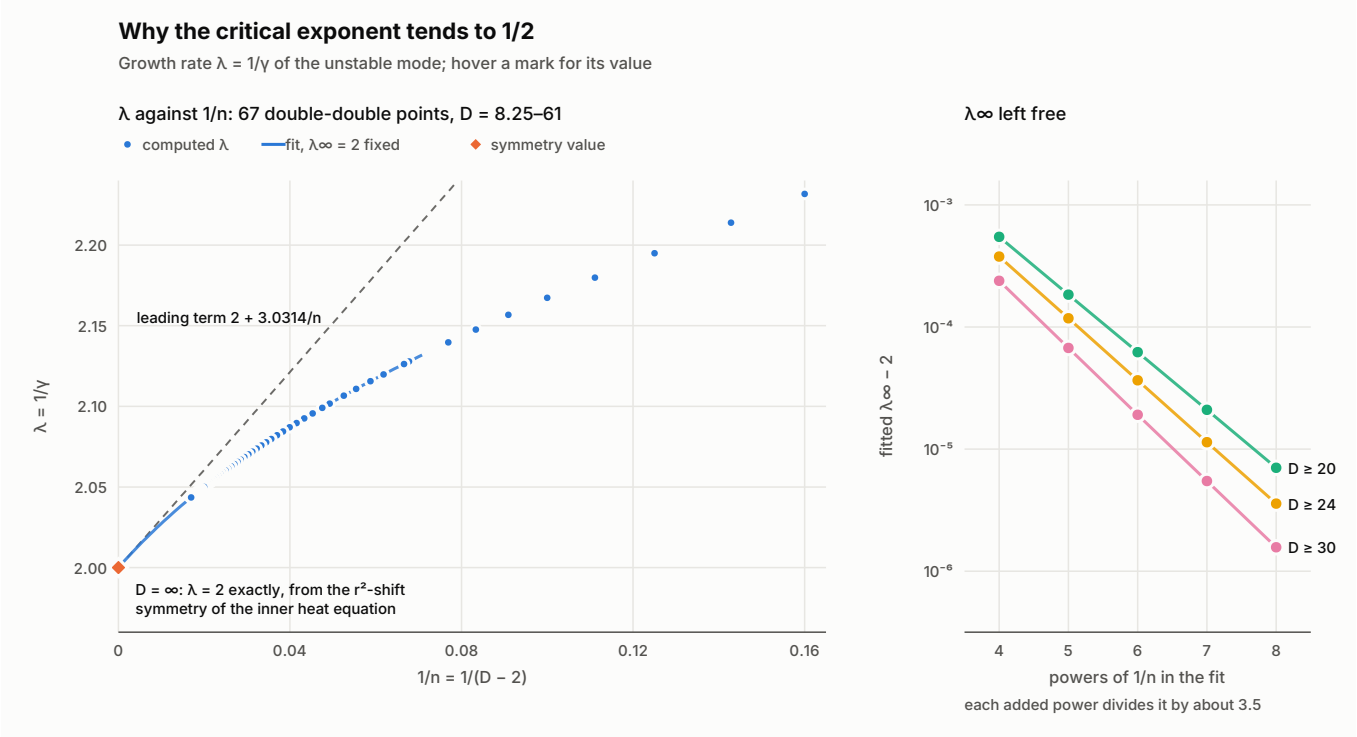


FIG. 6. Left: $\lambda = 1/\gamma$ against $1/(D-2)$ for 67 double-double points, with a fit fixing $\lambda_\infty = 2$ (solid, over its range $D \geq 16$) and its leading term $2 + 3.03/n$ (dashed); at $D = \infty$, $\lambda = 2$ is exact, from the r^2-shift symmetry of the inner heat equation. Right: with λ_∞ left free, the fitted $\lambda_\infty - 2$ falls about 3.5-fold per added power of $1/n$, for three fit ranges.

selects $\omega = (1.0000 \pm 0.0001)\,\omega_\infty$ untruncated and $0.99923\,\omega_\infty$ truncated to five harmonics. Open are: higher orders of that condition, which would give Δ_∞ (and possibly a closed form) from first principles to more digits; a proof of Conjecture 1 (a roadmap is in the repository's PROOF.md); the uniqueness of the $\lambda = 2$ mode and its $1/n$ correction; and the behaviour just below $D = 4$. We also assume, without checking it, that the DSS branch we follow remains the critical solution at large D, with a single unstable mode: our growth rate is a local root of the growth-mode determinant near its predicted value, other unstable modes were not searched for, and no collapse has been evolved above $D = 14$. If the branch acquired a second unstable mode, or a continuously self-similar solution [3] took over the threshold, $1/\lambda$ would not be the critical exponent. Other matter models are a natural test of Proposition 5: a centrifugal term of order n/r^2, as for scalar multiplets with angular momentum, wave maps or Yang–Mills, becomes an explicit $1/s$ in the inner equations and removes the symmetry. For the axion–dilaton, whose critical solution has no such term, the repository's computation of its continuously self-similar critical solution to $D = 80$ gives $\gamma \to 1/2$ (to about 3×10^{-6}), and the $D = \infty$ selection rule reproduces its frequency to 1.5×10^{-4}.

VII. HOW THIS WORK WAS DONE

N. M. chose the problem, set the direction at each step and ran the long computations on his own machine. The code, the computations, the derivations, the proofs and the text, including this paper, were produced by Claude, Anthropic's AI model, working as a set of coordinated agents. N. M. has not checked the mathematics himself, and no human expert has reviewed the work. Checks made: two independent solvers agree where they overlap; two machines with separate builds agree to $2\text{–}4 \times 10^{-16}$; every fit table reproduces from the data with the stated command; Propositions 1–3 and the inner equations were re-derived by separate agents (with sympy where possible); the inner Stokes multiplier was computed by two implementations that share no code; values at $D = 4$ and the maximum of Δ agree with the literature; an independent agent refereed the write-up; an independent review re-derived every input of the computer-assisted certificate, and deliberately broken versions of its final test fail as predicted; synthetic data with known limits test the error bar on Δ_∞; and every citation and novelty claim was checked against the published record. Corrections are welcome.

Data and code. All data, code and derivation scripts are at https://github.com/nmoinvaz/dss-residual-ker

nel, with a reproduction script (`./reproduce.sh`, which regenerates the fit tables and checks) and the double-double points in `hp/data/delta_dd.csv`.

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